Trend Pulse

Finding what the news is talking about, without a keyword list

Introduction to Big Data · Group project · First presentation

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Prototype v0.4 · November 2026



What is the news talking about right now?

Two easy answers, and why we did not take them

Keep a list of keywords and count them

Simple and tiny. But somebody has to write the list first.

Nobody had “scalia” on a list before February 13, 2016. A list only finds topics you already expected.

Count every word, forever

Finds everything. But the memory never stops growing.

By 2016 the exact count holds about 43,000 words. Most of them are from years ago and will never be hot again.

What we want: find new topics with no list, keep the memory small, and know exactly what we lose by keeping it small.

How a word gets its score

An exponentially decaying window with pruning (week 4), applied to every word and two-word phrase in a headline

1

A headline arrives

Split it into words and two-word phrases, such as “supreme court”.

2

Everything fades

All scores shrink a little every day. After 7 days a score is half of what it was.

3

New words get +1

Each word in the new headline gets one point. Hot words climb; quiet words sink.

4

Forget the small ones

A score below θ = 0.5 is deleted. This step is where the memory is saved.

What we can promise

A deleted word can lose at most `θ / (1 − θ) = 1 point`. So when the screen says 20.2, the true score is between 20.2 and 21.2. Nothing is lost silently.

How we check ourselves

Next to the small version we run an exact version that never deletes anything. It is only there to measure our errors. We also use DGIM as a helper, to count one chosen word over the last N articles.

Ten years of HuffPost headlines, replayed day by day

209,480

headlines, January 2012 to September 2022, in date order

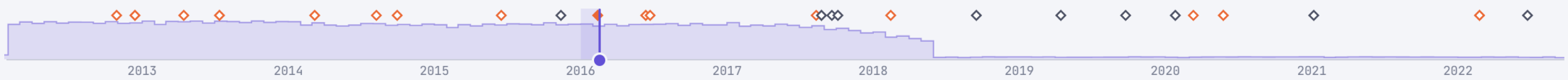
2 kinds

of keys: single words and two-word phrases

26 events

real-world events used as test cases, 10 of them chosen before the test

Headlines per month, and where the 26 test events sit on the timeline

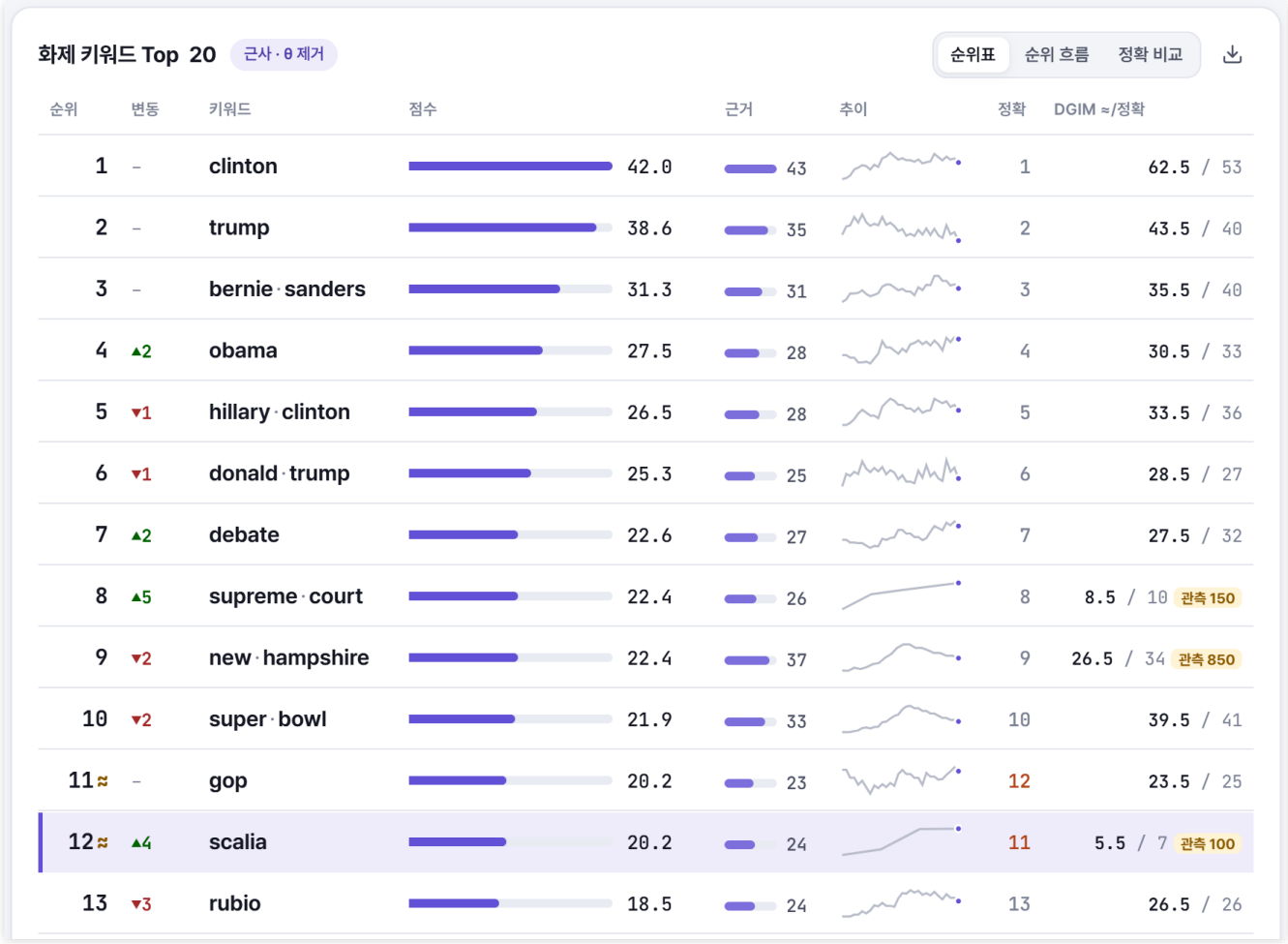


◇ **orange** = events we had from the start ◇ **grey** = events registered before this round the purple area is the number of headlines per month

Two things to keep in mind. The data has dates but no time of day, so this is a replay of the past, not live collection. And after 2018 the stream is thin, which changes some results later.

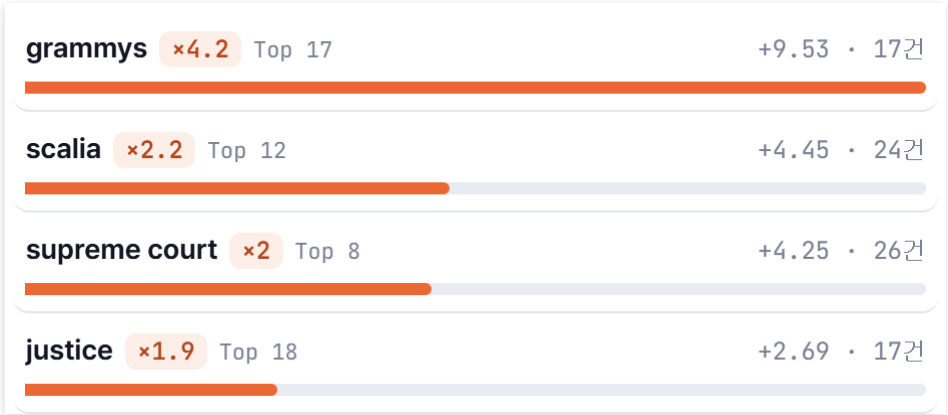
February 16, 2016: three days after Justice Scalia died

The v0.4 dashboard, paused at that date. Left: the Top 20. Right: words that are growing faster than usual.



순위	변동	키워드	점수	근거	추이	정확	DGIM ≈/정확
1	–	clinton	42.0	43		1	62.5 / 53
2	–	trump	38.6	35		2	43.5 / 40
3	–	bernie · sanders	31.3	31		3	35.5 / 40
4	▲2	obama	27.5	28		4	30.5 / 33
5	▼1	hillary · clinton	26.5	28		5	33.5 / 36
6	▼1	donald · trump	25.3	25		6	28.5 / 27
7	▲2	debate	22.6	27		7	27.5 / 32
8	▲5	supreme · court	22.4	26		8	8.5 / 10 관측 150
9	▼2	new · hampshire	22.4	37		9	26.5 / 34 관측 850
10	▼2	super · bowl	21.9	33		10	39.5 / 41
11 ≈	–	gop	20.2	23		12	23.5 / 25
12 ≈	▲4	scalia	20.2	24		11	5.5 / 7 관측 100
13	▼3	rubio	18.5	24		13	26.5 / 26

Bursts: words growing faster than usual



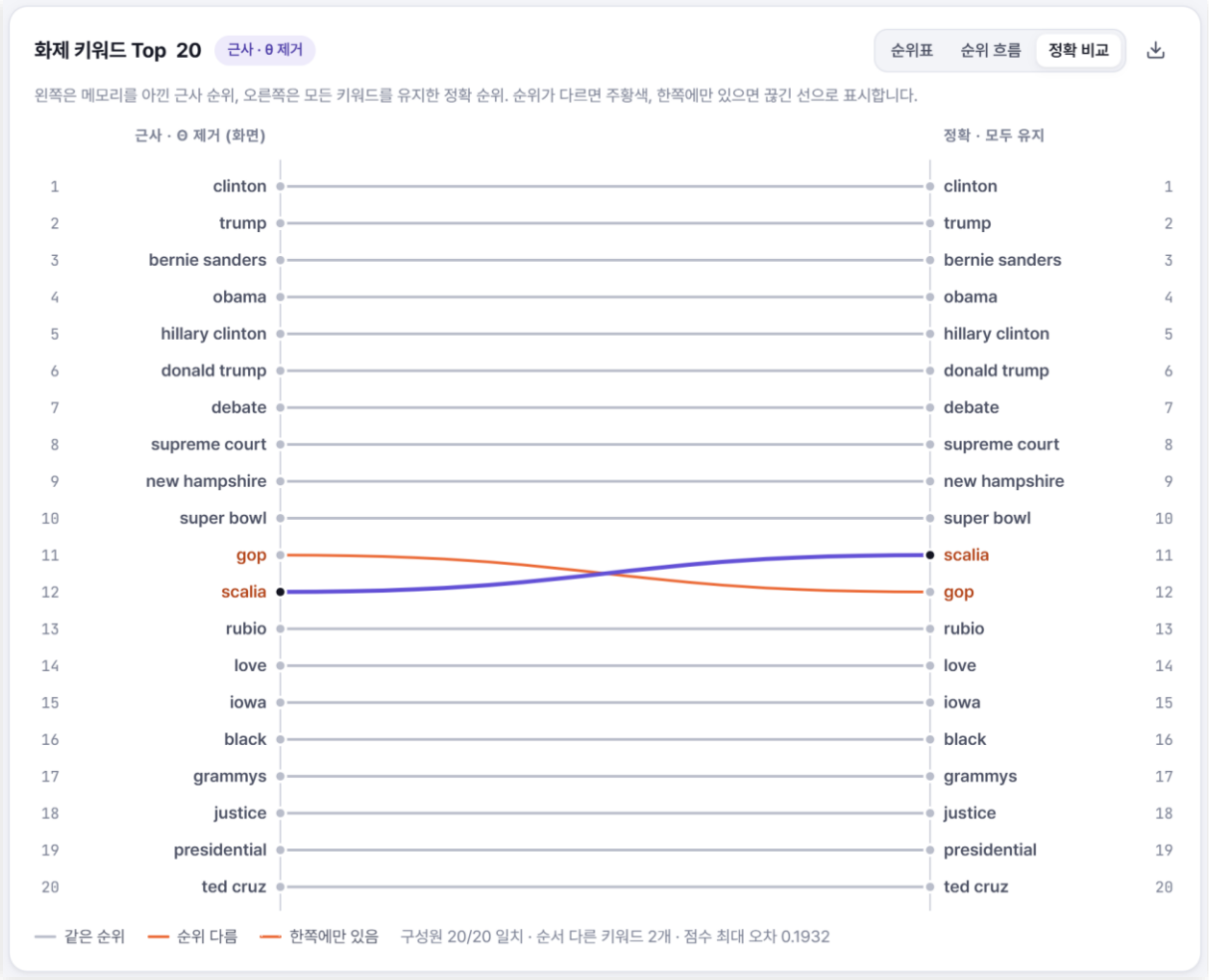
≈ “gop” and “scalia” have almost the same score. Saving memory may have swapped them, and it did: the exact ranks (정확) are 12 and 11.

근거 Evidence: how many headlines in the last 11 days contain the word. A rank with only one or two headlines is greyed out.

×2.2 “scalia” and “supreme court” are growing about two times faster than usual. Bursts often show an event before the Top 20 does.

What do we lose by saving memory?

Left: our ranking. Right: the exact version that never deletes. Same day as the demo.



−92.9%

stored keywords: 2,854 instead of 40,265 on average (θ = 0.5)

99.9%

of the time the Top 20 has the same members as the exact version

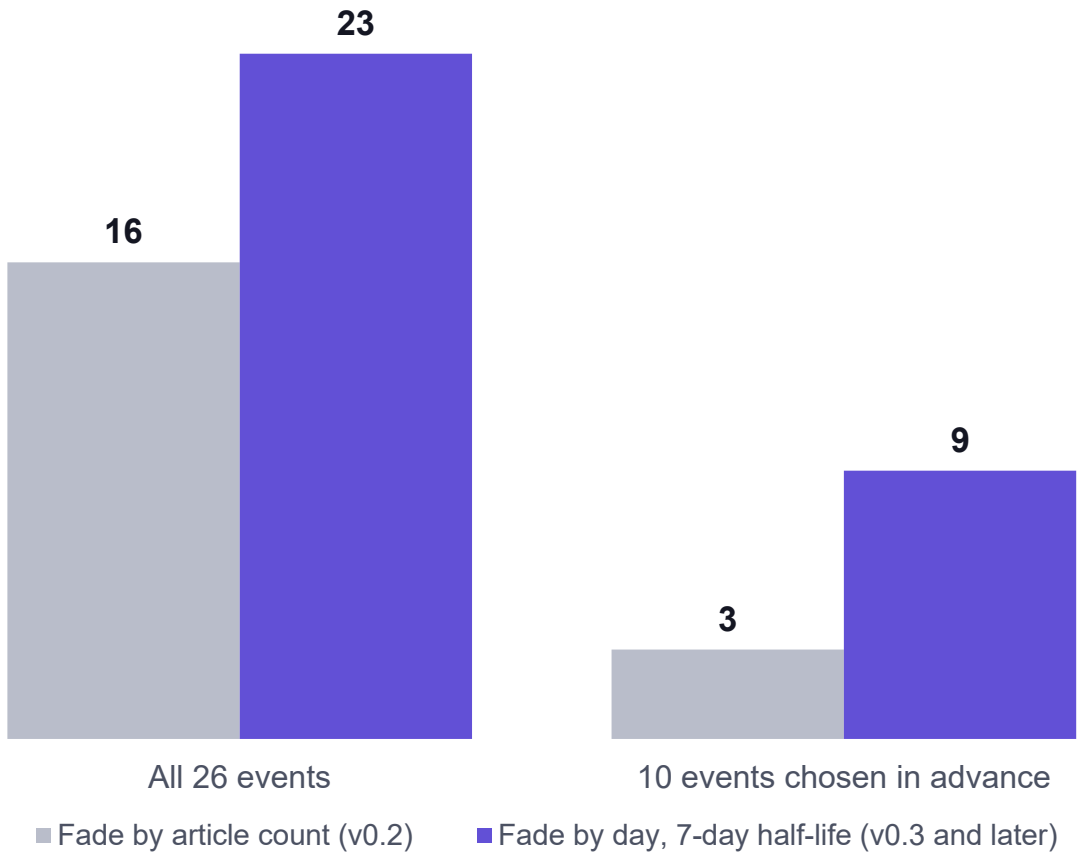
89%

of snapshots have exactly the same order. The rest differ by swaps like gop ↔ scalia

Swaps only happen between words with almost equal scores. We mark them with ≈, and in 37,905 neighbour pairs we missed none.

Does it find real events?

Found = the event's word enters the Top 20 within 7 days of the event



Why counting days works better

After 2018 HuffPost publishes about 5 headlines a day. If the score fades per 1,000 articles, that is months of old news still sitting in the Top 20. Fading per day lets a new event show up in 0 to 2 days.

Ten events were written down before we ran the test, so we could not pick the ones that happen to work.

Bursts are even faster

24 of 26 events appear in the burst list within 7 days, and in **17** cases they appear there before they reach the Top 20.

What is not solved yet

Open problems

Thin evidence after 2018. About 3 of 4 words in the Top 20 have two or fewer headlines in the last week. v0.4 now shows the evidence count and greys out thin rows, but the ranking itself is still thin.

Only HuffPost. The Top 20 is what one news site wrote about, not what the whole world cared about.

The explaining agent is untested with the real model. The “why is this trending?” assistant works in offline mode and checks every article it cites, but we have not run it with the real Solar API yet.

Until the second presentation

- Run the agent with the real API and grade its answers as a team.
- Test on random dates, not only on famous events, so the test is fairer.
- Try phrases of three or more words, such as “black lives matter”.
- Tune the burst detector for the thin years after 2018.
- Decide together: is “two headlines” the right line for thin evidence?

The algorithm and the numbers did not change in v0.4. This round was about the dashboard: timeline, evidence column, exact comparison, keyword details.



Thank you

Small memory, new topics, and we can say exactly what we got wrong.

Code, experiments and this deck: `github.com/legendkimofficial/TrendPulse`

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